

$$E[X] = (1-p) [1 + E[X]] + p$$

$$\Rightarrow E[X] [1 - (1-p)] = 1-p+p=1$$

$$\Rightarrow \boxed{E[X] = \frac{1}{p}}$$

$$E[X^2] = E[X^2|T_1] (1-p) + \underbrace{E[X^2|H_1]}_{=1} p$$

$$= E[X^2|T_1] (1-p) + p$$

$$E[X^2|T_1] = E[(1+X)^2]$$

H_1 : heads in first toss

$T_1: H_1^c$

$$E[X] = E[X|H_1]P(H_1) + E[X|T_1]P(T_1)$$

$X \sim \text{Geometric}(p)$

$$= 1 \times p + E[(1+X)](1-p)$$

$$E[X] = p + (1-p)[1 + E[X]]$$

$$pE[X] = 1$$

$$E[X^2] = E[X^2|H_1]P(H_1) + E[X^2|T_1]P(T_1)$$

$$= p \times 1 + (1-p)E[(1+X)^2]$$

$$E[X^2] =$$

$$\Rightarrow pE[X^2]$$

$$E[$$

$$\text{Var}(X)$$



$$\begin{aligned} E[X^2] &= p + (1-p) [1 + 2E[X] + E[X^2]] \\ &= 1 + 2(1-p) \times \frac{1}{p} + (1-p)E[X^2] \end{aligned}$$

$$\Rightarrow pE[X^2] = 1 + \frac{2(1-p)}{p} = \frac{2-p}{p}$$

$$E[X^2] = \frac{2-p}{p^2}$$

$$\text{Var}(X) = E[X^2] - (E[X])^2$$

$$= \frac{2-p}{p^2} - \frac{1}{p^2}$$

$$= \frac{1-p}{p^2}$$

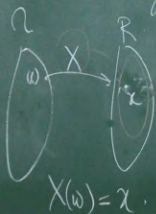
Continuous R.V.s.

Any random X is said to be continuous if its CDF $F_X(x)$

can be represented as

$$F_X(x) = \int_{-\infty}^x f_X(t) dt$$

for some function $f_X(t)$ referred to as probability density function (p.d.f.).



CDF of any R.V. X is defined as $\rightarrow$ monotonic, right continuous

$$F_X(x^+) = F_X(x)$$

$$\lim_{x \rightarrow -\infty} F_X(x) = 0$$

$$\lim_{x \rightarrow \infty} F_X(x) = 1$$

$$F_X(x) = P(X \leq x)$$

$$= P(\{\omega: X(\omega) \leq x\})$$

$X: (-\infty, \infty)$

Properties of pdf, Continuous R.V.s.

$$① f_X(x) \geq 0 \quad \forall x \in \mathbb{R}$$

$$② \int_{x \in \mathbb{R}} f_X(x) dx = 1$$

$$③ \int_a^b f_X(x) dx = P(a < X \leq b)$$

$$④ P(X=x) = 0$$

$$⑤ \lim_{h \rightarrow 0^+} F_X(x+h) = \lim_{h \rightarrow 0^+} \overbrace{F_X(x-h)}^{F_X(x)} = F_X(x)$$

$$⑤ \Rightarrow ④$$

$$\Omega = [0, 1]$$

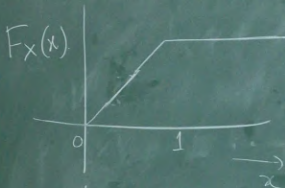
$$P((a,b)) = P(X(a) = w)$$

$$P(X=x) = F_X(x) - F_X(x^-) = 0$$

Example. $X(\omega) = \omega \quad \omega \in [0, 1]$

$$F_X(x) = P(X \leq x)$$

$$= \begin{cases} 0 & x \leq 0 \\ x & 0 \leq x \leq 1 \\ 1 & x > 1 \end{cases}$$



If $F_X(x)$ is differentiable,

then $f_X(x) = \frac{\partial F_X(x)}{\partial x}$ is a valid pdf for RV X

a) $f_X(x) \geq 0$

$$f_X(x) = \lim_{h \rightarrow 0} \frac{F_X(x+h) - F_X(x)}{h}$$

if $h > 0$ $F_X(x+h) \geq F_X(x)$ as CDF is monotonic

$$\Rightarrow \frac{F_X(x+h) - F_X(x)}{h} \geq 0$$

$h < 0$ $F_X(x+h) \leq F_X(x)$

$$\frac{F_X(x+h) - F_X(x)}{h} \geq 0 \Rightarrow f_X(x) \geq 0$$

$$\textcircled{b} \int_{-\infty}^a f_X(x) dx = \lim_{x \rightarrow \infty} F_X(x) = 1$$

$$\begin{aligned} \textcircled{c} \int_a^b f_X(x) dx &= \int_{-\infty}^b f_X(x) dx - \int_{-\infty}^a f_X(x) dx \\ &= F_X(b) - F_X(a) \\ &= P(a < X \leq b) \end{aligned}$$

$a \leq b$

$$A = \{X \leq a\}, B = \{X \leq b\}$$

$$A \subseteq B$$

$$\begin{aligned} P(A^c \cap B) &= P(B) - P(A \cap B) \\ &= P(B) - P(A) \end{aligned}$$

Enough to show left continuity, given that X is a continuous RV

i.e., $\exists a$ for $f_X(x)$ s.t.

$$F_X(x) = \int_a^x f_X(t) dt$$

$$\lim_{h \rightarrow 0^+} F_X(x) - F_X(x-h) = 0.$$

$$F_X(x) - F_X(x-h) = \int_{x-h}^x f_X(t) dt.$$

$$\lim_{h \rightarrow 0^+} F_X(x) - F_X(x-h) = \lim_{h \rightarrow 0^+} \int_{x-h}^x f_X(t) dt = 0.$$

$$\Rightarrow \lim_{h \rightarrow 0} (F_X(x) - F_X(x-h)) = 0$$

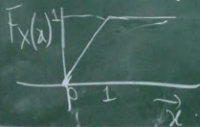
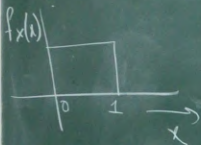
$$\Rightarrow F_X(x^-) = F_X(x).$$

Examples

(a) $X \sim \text{Uniform}([0, 1])$.

$$f_X(x) = \begin{cases} 1 & x \in [0, 1] \\ 0 & \text{otherwise} \end{cases}$$

$$\int_0^1 f_X(x) dx = \int_0^1 1 dx = 1$$



(b) $X \sim \text{Uniform}([a, b])$.

$$f_X(x) = \begin{cases} c & x \in [a, b] \\ 0 & \text{otherwise} \end{cases}$$

find c

$$\int_a^b f_X(x) dx = 1 \Rightarrow c(b-a) = 1$$

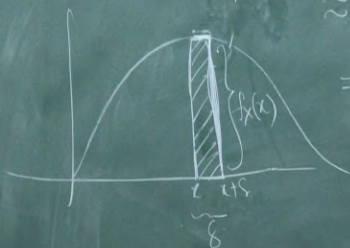
$$(2) f_X(x) = \begin{cases} c/\sqrt{x} & x \in (0, 1] \\ 0 & \text{otherwise} \end{cases}$$

Find c

$$\begin{aligned} \int_{-\infty}^{\infty} f_X(x) dx &= \int_0^1 \frac{c}{\sqrt{x}} dx \\ &= c \left. \frac{x^{-1/2+1}}{-1/2+1} \right|_0^1 \\ &= 2c = 1 \\ \Rightarrow c &= 1/2 \end{aligned}$$

Intuition for the pdf $f_X(x)$

$$\begin{aligned} P(x \leq X \leq x+\delta) &= \int_x^{x+\delta} f_X(t) dt \\ &\approx f_X(x) \int_x^{x+\delta} dt \\ &= f_X(x) \delta \end{aligned}$$



Density function at x is
the probability of R.V at
vicinity of x per the interval
size ' δ '

Expectation of continuous R.V.s.

$$E[X] = \int_{-\infty}^{\infty} x \underbrace{f_X(x)}_{\text{approximates the probability in vicinity of } x} dx$$

approximates
the summation.

approximates the
probability in
vicinity of x .

$$\text{If } \int_{-\infty}^{\infty} |x| f_X(x) dx < \infty.$$

then $E[X]$ is well defined